Report: Toy Implementation of Phrase-Based and Neural Unsupervised Machine Translation

Based on Lample et al., "Phrase-Based & Neural Unsupervised Machine Translation"

Nooshin Behrooz - July 2026

1. Introduction

The paper I studied is "Phrase-Based & Neural Unsupervised Machine Translation" by Lample et al. The main question of the paper is simple but important: can we build a machine translation system without using parallel sentence pairs?

Normally, machine translation systems need parallel corpora. For example, the model needs an English sentence and its correct translation in another language. The model learns by comparing these sentence pairs and learning word, phrase, and sentence alignments. However, many language pairs do not have enough parallel data, especially low-resource languages.

In this project, I first studied the main ideas of the paper, and then I implemented two simplified toy systems for English-Italian translation: one Toy NMT model and one Toy PBSMT model. The goal was not to reproduce the exact results of the original paper, because the paper uses much larger datasets and stronger models. My goal was to understand the structure of the method and show the same logic in a smaller experiment that can run in Google Colab.

2. Main Idea of the Paper

The paper shows that unsupervised machine translation can work by combining three main principles: initialization, language modeling, and iterative back-translation. These three ideas are used both in the neural model and in the phrase-based model.

2.1 Initialization

Because there is no parallel data, the system needs a weak starting bridge between the two languages. In NMT, this is done using shared embeddings from monolingual corpora. In PBSMT, this is done by building an initial phrase table from a bilingual dictionary inferred from cross-lingual embeddings. This step is important because if the model starts from complete randomness, later back-translation may not be enough to fix it.

2.2 Language Modeling

After initialization, the translations are still noisy and unnatural. Language modeling helps the system learn what a correct sentence looks like in each language. In NMT, this is done with a denoising autoencoder: the model receives a corrupted sentence and learns to reconstruct the clean sentence. In PBSMT, a statistical n-gram language model scores translation candidates and prefers more fluent sentences.

2.3 Iterative Back-Translation

Back-translation is the step that creates pseudo-parallel data automatically. The model translates monolingual sentences into the other language, and then these synthetic sentence pairs are used for training. This is repeated many times. Early translations are noisy, but later translations can become better and improve the model.

3. Paper Results

In Table 2, the paper compares its proposed unsupervised models with previous unsupervised MT methods. The proposed models strongly outperform previous approaches. For English to French, the previous best unsupervised result was about 17 BLEU, while the paper reports much higher scores:

Model	BLEU for en-fr
Previous best unsupervised MT	17.0
Paper NMT Transformer	25.1
Paper PBSMT	28.1
Paper PBSMT + NMT	27.6

A second important result is that PBSMT is surprisingly strong. In many cases, the PBSMT model is stronger than the NMT model, although neural models are often expected to be stronger. The hybrid system, especially PBSMT + NMT, gives the best overall big-picture performance, especially for harder language pairs.

In Table 3, the paper also shows that increasing the number of back-translation iterations improves PBSMT. This supports the idea that iterative back-translation is one of the key reasons for the success of the system.

Figure 2 compares supervised and unsupervised approaches. The supervised models improve when more parallel sentence pairs are used. The unsupervised models use zero parallel sentences, so their result is fixed. The main message is that unsupervised MT is especially useful when parallel data is scarce, noisy, or unavailable. However, when a supervised model has a very large parallel corpus, supervised MT becomes stronger.

4. My Toy NMT Implementation

For my first implementation, I built a simplified Toy NMT model for English-Italian translation. I used the Helsinki-NLP OPUS Books English-Italian dataset, but English and Italian training sentences were taken from different ranges, so they were used as non-parallel monolingual corpora. Parallel references were used only for BLEU evaluation.

The model is a small Transformer encoder-decoder model. It uses one shared vocabulary and one shared embedding layer for both English and Italian. This is a simplification of the paper. In the paper, the NMT model uses joint BPE and stronger shared embeddings, while my toy model learns a simple shared vocabulary and embedding layer from scratch.

4.1 Toy NMT Training Steps

- Monolingual autoencoder warm-up: learns to reconstruct English sentences from English input and Italian sentences from Italian input, without using parallel sentence pairs.
- Denoising autoencoder: corrupts English and Italian sentences and trains the model to reconstruct the clean version.
- Joint denoising + light back-translation: combines denoising losses and back-translation losses in one joint update.

I evaluated the Toy NMT model with BLEU on 50, 100, 250, and 500 sentences. BLEU is an automatic evaluation score that compares n-gram overlap between the machine translation and the human reference translation.

Number of Sentences	Non-Parallel Toy NMT BLEU
50	0.07
100	0.12
250	0.08
500	0.06

The BLEU scores are low, but this is expected because the model is very small, the training is light, and the training corpora are non-parallel. The purpose was mainly to follow the structure of the paper: initialization, denoising, and back-translation.

5. My Toy PBSMT Implementation

For the second implementation, I built a simplified Toy PBSMT model for English-Italian translation. This model also follows the three-level structure of the paper.

5.1 Toy PBSMT Training Steps

- Initialization: trains separate Word2Vec embeddings for English and Italian, aligns the embedding spaces, retrieves nearest neighbors, and builds an initial phrase table.
- Language modeling and decoding: trains a bigram language model and uses it to score fluent Italian translation candidates.
- Bidirectional back-translation: creates synthetic English-Italian and Italian-English pairs and updates the phrase table.

The Toy PBSMT model has a clear connection to Information Retrieval. In the initialization step, it retrieves nearest neighbors in the embedding space using similarity. In decoding, it generates candidates, scores them, and selects the best translation.

I evaluated the final Toy PBSMT system after bidirectional back-translation using BLEU on the same test sizes:

Number of Sentences	Toy PBSMT BLEU (After Bidirectional BT)
50	0.19
100	0.17
250	0.19
500	0.17

The Toy PBSMT scores after bidirectional back-translation are still higher than the Toy NMT scores. This is similar to the paper’s result, where PBSMT is surprisingly strong and often better than NMT in the unsupervised low-resource setting.

However, in my toy experiment, back-translation did not always improve the PBSMT result. The reason is that the synthetic translations were still noisy. If noisy synthetic data is added to the phrase table, it can weaken the model instead of improving it.

6. Comparison Between Toy NMT and Toy PBSMT

The main comparison in my implementation is between Toy NMT and Toy PBSMT. With the updated non-parallel setting, the Toy NMT BLEU scores are very low, while the Toy PBSMT BLEU scores after bidirectional back-translation are slightly higher. Therefore, my Toy PBSMT still performs better than my Toy NMT in this toy experiment.

Number of Sentences	Non-Parallel Toy NMT BLEU	Toy PBSMT BLEU (After BT)
50	0.07	0.19
100	0.12	0.17
250	0.08	0.19
500	0.06	0.17

This result is consistent with the paper's big-picture observation. PBSMT can work well in this setting because phrase tables and language models are easier to bootstrap from weak bilingual signals. NMT has more parameters and usually needs more data to learn good representations.

7. Conclusion

In this project, I studied the paper "Phrase-Based & Neural Unsupervised Machine Translation" and implemented two simplified English-Italian toy systems inspired by it.

The paper shows that unsupervised MT can work by combining initialization, language modeling, and iterative back-translation. It also shows that PBSMT is surprisingly strong and that combining PBSMT with NMT gives the best overall performance.

In my implementation, I built a Toy NMT and a Toy PBSMT. The updated Toy NMT used non-parallel monolingual corpora, a shared vocabulary and embedding layer, monolingual autoencoder warm-up, and joint denoising plus light back-translation. The Toy PBSMT followed the idea of phrase table initialization, language modeling, decoding, and bidirectional back-translation.

The final updated result was that Toy PBSMT achieved higher BLEU than Toy NMT, even though both scores were low in the non-parallel toy setting. This is consistent with the paper's observation that PBSMT can outperform NMT in low-resource unsupervised settings.

Overall, this project helped me understand how machine translation can be trained without large parallel corpora, and how ideas such as embedding alignment, language modeling, candidate ranking, and back-translation can work together in unsupervised MT.

Reference

Lample, G., Ott, M., Conneau, A., Denoyer, L., & Ranzato, M. (2018). Phrase-Based & Neural Unsupervised Machine Translation. arXiv:1804.07755v2.